

Flow Control in Computer Networks

Flow control in computer networks is a technique that ensures the sender does not overwhelm the receiver by sending data faster than it can be processed. It helps match the data transmission rate of the sender and the receiver.

Key Points

- Purpose: Prevent buffer overflow at the receiver.
- Layers: Works at Data Link Layer and Transport Layer.
- Idea: Match sender's speed with receiver's processing capability.

Techniques of Flow Control

1. Stop-and-Wait Flow Control

- Sender sends one frame at a time and waits for acknowledgment (ACK).
- Once ACK is received, the next frame is sent.
- Simple, but inefficient for long-distance or high-speed links.

2. Sliding Window Flow Control

- Sender can send multiple frames before needing ACK, depending on window size.
- Receiver stores frames in a buffer window.
- Window slides forward as ACKs arrive, allowing new frames to be sent.

Example (Window Size = 4, Frames 0-7):

- Initial Window: [0][1][2][3] → Can send now
- After ACK(0): [1][2][3][4] → Frame 4 is new
- After ACK(1): [2][3][4][5] → Frame 5 is new
- If Cumulative ACK(4): [4][5][6][7] → All are new to send

Types of Sliding Window Protocols

- Go-Back-N ARQ → If an error occurs, all frames from the erroneous one onward are retransmitted.
- Selective Repeat ARQ → Only the erroneous/lost frame is retransmitted; others are buffered.

Flow Control vs Error Control

- Flow Control: Prevents sender from overwhelming the receiver.
- Error Control: Ensures reliable delivery (detects and retransmits lost/damaged frames).

Summary

- Flow control ensures smooth communication between sender and receiver.
- Stop-and-Wait is simple but inefficient.

- Sliding Window is more efficient and widely used.
- Cumulative ACKs allow the window to slide multiple steps at once.
- Efficient flow control is essential for reliable and fast networks.